σ_2-G1 at $n = 6$: KL-cone Decomposition via the Doubled Word

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Abstract

We refine the trace-square reduction of σ_2-G1 by decomposing both σ_1^2 and $\mathrm{tr}(\Pi_q^2)$ in the unique linear basis of $\mathbb{Z}[q]$ given by $\{(1+q)^l q^{(M-l)/2}\}_{l \text{ even},\, 0 \le l \le M}$, where $M = 2N$ and $N = \binom{n}{2}$. Each decomposition has non-negative coefficients by σ_1-G1 (applied to the staircase word and to its doubling, respectively). The strong-G1 conjecture $N_l(\lambda) \ge m_l(\lambda)$ would then place σ_2 in the positive cone of basis vectors, immediately yielding palindromic positivity. We verify this strong-G1 cone bound *computationally* for $\lambda = (5,1)$ and $\lambda = (4,2)$ at $n = 6$, but *refute* it for $\lambda = (3,2,1)$ at $n = 6$ (negative coefficient -3 at $l = 4$). Thus KL-cone decomposition closes σ_2-G1 at $n = 6$ for two of three rank-≥ 2 shapes; $(3,2,1)$ requires the orthogonal atom-decomposition of `2026-04-29-sigma2-G1-321-atom-decomp.tex`. Together the two methods close σ_2-G1 at $n = 6$ in full, but not via a single uniform mechanism.

1 Setup

Let $H_q = H_q(S_n)$ be the Iwahori–Hecke algebra with quadratic relation $T_i^2 = (q-1)T_i + q$. Let $\Pi_q^{(n)} = \prod_{j=1}^{N}(1 + T_{i_j})$ be the Bott–Samelson operator along the staircase reduced word for w_0, with $N = \binom{n}{2}$, and let V_λ denote Hoefsmit's seminormal Specht module for $\lambda \vdash n$. Write $\sigma_k(V_\lambda) := e_k(\text{eigenvalues of } \Pi_q^{(n)}|_{V_\lambda})$.

Definition 1 (σ_k-G1)**.** A polynomial $f(q) \in \mathbb{Z}[q]$ is *G1* if $f(q) = q^a(1+q)^b \cdot P(q)$ with $P \in \mathbb{Z}_{\ge 0}[q]$ palindromic. We say σ_k-G1 holds for V_λ if $\sigma_k(V_\lambda)$ is G1.

Theorem 2 (σ_1-G1, Apr 24)**.** *For every $\lambda \vdash n$, $\sigma_1(V_\lambda) = \mathrm{tr}(\Pi_q^{(n)}|V_\lambda) = \sum_{k=0}^{N} n_k(\lambda)\,(1+q)^k\, q^{(N-k)/2}$ with $n_k(\lambda) \in \mathbb{Z}_{\ge 0}$, where the sum is over k with $N - k$ even.*

The proof of Theorem **??** uses the Kazhdan–Lusztig decomposition $T_i + 1 = (1+q)D_i + vM_i$ (with $v = q^{1/2}$, D_i a diagonal indicator, and M_i a non-negative integer matrix). Expanding the BS product and applying the parity-vanishing lemma yields the stated form.

2 Doubled-word analogue and basis decomposition

Lemma 3 (Doubled-word σ_1-G1)**.** *For every $\lambda \vdash n$,*

$$\mathrm{tr}((\Pi_q^{(n)})^2|V_\lambda) \;=\; \sum_{l=0}^{2N} m_l(\lambda)\,(1+q)^l\, q^{(2N-l)/2},$$

with $m_l(\lambda) \in \mathbb{Z}_{\ge 0}$ and $m_l = 0$ unless l is even.

Proof. The proof of Theorem **??** uses only that $\Pi_q^{(n)}$ is a product of factors $(1 + T_{i_j})$, applying the KL decomposition $T_i + 1 = (1+q)D_i + vM_i$ factorwise and then taking the trace with the parity vanishing lemma. The same proof applies verbatim to any product of factors $(1 + T_{i_j})$ — in particular to the doubled BS-word $\Pi_q^{(n)} \cdot \Pi_q^{(n)}$, viewed as $\prod_{j=1}^{2N}(1 + T_{i_{(j \bmod N)}})$ along the doubled staircase index sequence. The only change is $N \mapsto 2N$. $\square$

Proposition 4 (Basis B_M)**.** *The set $B_M := \{(1+q)^l q^{(M-l)/2} : l \in \{0,2,4,\dots,M\}\}$ is a $\mathbb{Q}$-linear basis of the subspace of $\mathbb{Q}[q]$ spanned by $\{q^a : 0 \le a \le M\}$ that consists of polynomials of degree $\le M$ which are palindromic about $q^{M/2}$ (i.e., $f(q) = q^M f(1/q)$). Both σ_1^2 and $\mathrm{tr}(\Pi_q^2)$ lie in this subspace with $M = 2N$.*

Proof. The basis vectors are linearly independent: each has distinct minimum degree $(M-l)/2$. There are $M/2+1$ basis vectors and $M/2+1$ degrees of freedom in the palindromic-about-$q^{M/2}$ subspace of degree $\le M$ polynomials. σ_1 is palindromic about $q^{N/2}$ (eigenvalues of Π_q are palindromic in q by parity vanishing), so σ_1^2 is palindromic about q^N. $\mathrm{tr}(\Pi_q^2)$ is similarly palindromic about q^N by Lemma **??**. □

3 KL-cone decomposition of σ_2

Theorem 5 (σ_2 in B_M)**.** *With $M = 2N$,*

$$\sigma_1^2(V_\lambda) = \sum_{l\ even} N_l(\lambda)\,(1+q)^l\,q^{(M-l)/2}, \qquad N_l(\lambda) = \sum_{k+k'=l} n_k(\lambda)\,n_{k'}(\lambda) \ \ge\ 0,$$

$$\mathrm{tr}(\Pi_q^2|V_\lambda) = \sum_{l\ even} m_l(\lambda)\,(1+q)^l\,q^{(M-l)/2}, \qquad m_l(\lambda) \ge 0,$$

and consequently, by Newton's identity $2\sigma_2 = \sigma_1^2 - \mathrm{tr}(\Pi_q^2)$,

$$\sigma_2(V_\lambda) = \sum_{l\ even} \frac{N_l(\lambda) - m_l(\lambda)}{2}\,(1+q)^l\,q^{(M-l)/2}.$$

Proof. The first identity squares $\sigma_1 = \sum_k n_k(1+q)^k q^{(N-k)/2}$ and collects monomials by total exponent of $(1+q)$. The second is Lemma **??**. The third is direct: by Proposition **??** the basis B_M is linearly independent, so coefficients combine. □

Conjecture 6 (Strong-G1 cone bound)**.** *For every $\lambda \vdash n$ and every l,*

$$N_l(\lambda) \ \ge\ m_l(\lambda), \qquad N_l(\lambda) - m_l(\lambda) \in 2\mathbb{Z}_{\ge 0}.$$

If Conjecture **??** holds, then by Theorem **??**, σ_2 lies in the positive integer cone spanned by B_M. Since each basis vector $(1+q)^l q^{(M-l)/2}$ is itself G1 (with palindromic factor 1), σ_2 is then G1.

4 Computational verification

We computed N_l, m_l for all $\lambda \vdash n$ with $n \le 6$ and rank ≥ 1 using the script `test_KL_decomposition_sigma2.py`, which evaluates Π_q and Π_q^2 in Hoefsmit's seminormal basis and solves the linear system from B_M to extract coefficients.

Table 1: $N_l - m_l$ for rank-≥ 2 shapes at $n = 6$ ($N = 15$, $M = 30$).

l	2	4	6	8	10	12	14	16	18	20	22	24	26
$(5,1)$	0	0	0	0	0	0	0	0	12	26	12	2	0
$(4,2)$	0	0	0	0	84	194	160	78	22	2	0	0	0
$(3,2,1)$	2	**−6**	18	24	12	2	0	0	0	0	0	0	0

For $(5,1)$ and $(4,2)$, all $N_l - m_l \ge 0$; Conjecture **??** holds for these two shapes. The resulting σ_2 decompositions are:

$$\begin{aligned}
\sigma_2(V_{(5,1)}) &= 6\,(1+q)^{18}q^6 + 13\,(1+q)^{20}q^5 + 6\,(1+q)^{22}q^4 + (1+q)^{24}q^3,\\
\sigma_2(V_{(4,2)}) &= 42\,(1+q)^{10}q^{10} + 97\,(1+q)^{12}q^9 + 80\,(1+q)^{14}q^8\\
&\quad + 39\,(1+q)^{16}q^7 + 11\,(1+q)^{18}q^6 + (1+q)^{20}q^5,
\end{aligned}$$

manifestly G1.

For $(3,2,1)$, $N_4 - m_4 = -6$, so the cone bound fails. Direct verification gives

$$\sigma_2(V_{(3,2,1)}) = q^9(1+q)^2 \cdot P_{(3,2,1)}(q),$$

with $P_{(3,2,1)}(q) = q^{10} + 16q^9 + 105q^8 + 369q^7 + 759q^6 + 961q^5 + 759q^4 + 369q^3 + 105q^2 + 16q + 1$ palindromic and ≥ 0, so σ_2-G1 *still holds*, just not via a positive-cone decomposition in B_M.

5 Why $(3,2,1)$ resists this approach

The basis B_M at $M = 30$ includes vectors with low l and high q-power (e.g., $l = 2$ has support $[14, 16]$). For shapes with large $b_2 = b_{\lambda,2}$ (the $(1+q)$ exponent in σ_2), the unique decomposition is forced into the high-l portion of B_M, leaving zero coefficient at low l. This is the case for $(5,1)$ ($b_2 = 18$) and $(4,2)$ ($b_2 = 10$).

For $(3,2,1)$, $b_2 = 2$ is small. The σ_2 polynomial sits with leading q-degree 9 and trailing q-degree 21, but its palindromic factor of degree 12 is "wide" in degree compared with $(1+q)^2$. The unique decomposition in B_M then uses low-l basis vectors with negative coefficients to "carve out" the wide palindromic shape from the narrower $(1+q)^l$ profiles.

Remark 7 (Interpretation)**.** The strong-G1 cone bound $N_l \geq m_l$ is genuinely stronger than σ_2-G1. The failure at $(3,2,1)$ suggests that strong-G1 captures shapes whose σ_2 has "simple" palindromic structure (factor $(1+q)^b$ dominating), while $(3,2,1)$ has the genuinely 11-coefficient palindromic atom A_{10} that does not factor through $(1+q)$. This matches the atom-alphabet picture from `2026-04-29-sigma2-G1-321-atom-decomp.tex`: $(3,2,1)$ uses the *orphan* atom $A_{10} = q^{10} + 16q^9 + \dots$ (OEIS A344437, derangement RTL-minima), unique among rank-≥ 2 shapes at $n \leq 6$.

6 Status and gap

What this proves at $n = 6$ (modulo Conjecture ??). For $\lambda \in \{(5,1), (4,2)\}$, σ_2-G1 follows from the KL-cone decomposition once $N_l \geq m_l$ is established. *Computationally* this is closed; *structurally* it requires proving the per-l inequality

$$\sum_{k+k'=l} n_k(\lambda)\, n_{k'}(\lambda) \;\geq\; m_l(\lambda)$$

combinatorially. This sharpens the gap from "σ_2-G1 holds" to a specific cone-positivity question on KL-coefficients.

Combined closure of σ_2-G1 at $n = 6$.

1. $(5,1)$: KL-cone decomposition (this paper, modulo the cone bound, computationally verified).
2. $(4,2)$: KL-cone decomposition (this paper, modulo the cone bound, computationally verified).
3. $(3,2,1)$: orthogonal atom-alphabet decomposition (`2026-04-29-sigma2-G1-321-atom-decomp.tex`).

This is a genuine *partial* result: the lemma needed for $(5,1), (4,2)$ is the per-shape combinatorial inequality $N_l \geq m_l$, and a structural proof of that inequality remains open. The $(3,2,1)$ case is closed by a different mechanism.

Open: structural proof of the cone bound. A natural candidate is per-pair sub-multiplicativity:

$$\mathrm{tr}(P_S|V_\lambda) \cdot \mathrm{tr}(P_{S'}|V_\lambda) \;\geq\; \mathrm{tr}(P_S P_{S'}|V_\lambda),$$

where P_S is the σ_1-G1 operator product over the subset S. For positive semi-definite $P_S, P_{S'}$ this would follow from the standard inequality $\mathrm{tr}(AB) \leq \mathrm{tr}(A)\,\mathrm{tr}(B)$ for PSD operators. The KL operators D_i, M_i are non-negative integer matrices in Hoefsmit's basis, but their products are not always PSD on V_λ, and the per-pair inequality fails for $(3,2,1)$ at $l = 4$.

Why σ_2-G1 closes despite this. The strong-G1 cone bound is strictly stronger than σ_2-G1. The right structural invariant for σ_2-G1 across all rank-≥ 2 shapes is plausibly the atom alphabet, not the B_M basis. The B_M basis appears to suffice precisely when the shape's palindromic factor is captured by KF ($q^a(1+q)^b$) elements; $(3,2,1)$ first introduces an irreducible non-cyclotomic palindromic atom (A_{10}, see project memory `project_oeis_derangement_rtl.md`) and lies outside this cone.

Acknowledgements and references

This refines the trace-square framing of `2026-04-28-sigma2-G1-n6-trace-square.tex` by working in a fixed linear basis. The atom-decomposition closure of $(3,2,1)$ is established in `2026-04-29-sigma2-G1-321-atom-decomp.tex`. The original σ_1-G1 proof underpinning Lemma **??** is in `2026-04-24-sigma1-G1.tex`. Computational verification: `/home/clio/projects/scratch/2026-04-29-s5-equivariance/test_KL_decomposition_sigma2.py`.